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Estimates of flow direction for calc-alkaline welded tuffs and paleomagnetic data reliability from anisotropy of magnetic susceptibility measurements: central San Juan Mountains, southwest Colorado

Brooks B. Ellwood

Department of Geology, University of Georgia, Athens, GA 30602 (U.S.A.)

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Flow directions are estimated from the measurement of the magnetic fabric of 106 samples, collected at 18 sites in four welded tuff units in the central San Juan Mountains of southern Colorado. The estimates assume that the tuffs generally flowed directly away from the extrusive vents and that the lineations of magnetic grains within the tuffs represent the flow direction at individual sites. Errors in the estimation may arise from topographic variation, rheomorphism (post-emplacement mass flow) within the tuff, and other factors. Magnetic lineation is defined as the site mean anisotropy of magnetic susceptibility maximum azimuth. A test on the flow directions for individual units is based on the projection of lineation azimuths and their intersection within or near the known source caldera for the tuff. This test is positive for the four units examined. Paleomagnetic results for these tuffs are probably reliable indicators of the geomagnetic field direction in southwest Colorado, during the time (28.2–26.5 Ma) of emplacement.

1. Introduction

The question that this research was designed to answer was: Can reasonable estimates of flow direction be obtained from anisotropy of magnetic susceptibility (AMS) measurements of tuff samples? To investigate this, several large volume tuffs were sampled for which careful geologic mapping had identified a known source caldera. Ideally, the magnetic lineation directions for several sites could be protected to a point of intersection that would lie in the source caldera. The central San Juan Mountains ash-flow tuff sequence in southern Colorado (Fig. 1) [1–3] was chosen for this work. This choice was based on the fact that the geology is well known, and that J.A. Whitney and J.C. Stormer, Jr. of the University of Georgia were working in the central San Juan Mountains.

This research provides some additional criteria, besides those reported by Geissman [4], that can be used in evaluating the reliability of paleomagnetic directions from tuffs. Furthermore, the central San Juan volcanic sequence is an excellent choice for such an analysis, because a number of paleomagnetic studies have been reported for these units [5,6] and the source of these tuffs has been well established [1,2].

This paper presents the following: (1) estimates of flow direction for four welded ash-flow tuffs in the central San Juan Mountains of southern Colorado; (2) identification of possible secondary mass flow (rheomorphism [7]) in these tuffs using magnetic fabric data; (3) evaluation of the effect of rheomorphism on some paleomagnetic directions; and (4) the induced and remanent magnetic varia-

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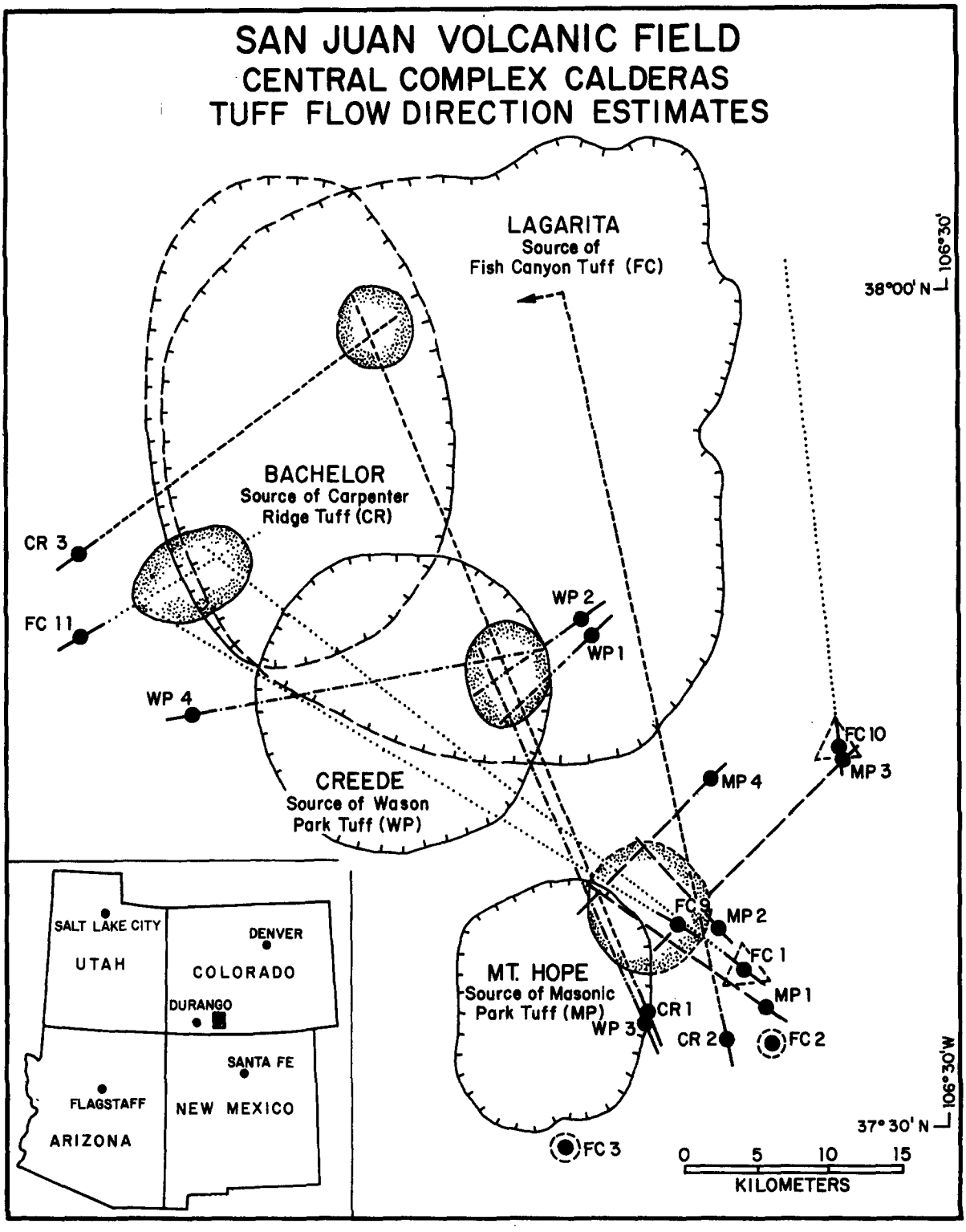


Fig. 1. Central San Juan volcanic field, the location of sites where samples were obtained, and the calderas from which the tuffs originated. Dashed circles represent sites for which no direction can be obtained. Dashed triangles delimit sites which appear to exhibit rheomorphic secondary flow [7], see text. Lines in Fig. 1 represent flow direction estimates, see text. FC = Fish Canyon tuff samples (dotted lines), FC-9 represents samples from sites FC-4 to FC-9, obtained from a ~250 m vertical section through the tuff; WP = Wasen Park tuff (dashed-dotted lines); CR = Carpenter Ridge tuff (short dash lines); MP = Masonic Park tuff (long dash lines). Note that the Creede cauldera is the *suspected* area from which the Wason Park tuff originated.

bility for a ~250 m stratigraphic section through a single tuff unit.

2. Previous work

2.1. Geological setting

Tertiary volcanism resulted in a sequence of calc-alkaline, welded ash-flow tuffs in the central San Juan Mountains of southern Colorado [1] (Fig. 1). Caldera locations, stratigraphy, tuff outcrop pattern, and whole-rock and mineral K-Ar ages have been reported [1–3]. They suggest that these tuffs originated from a large composite batholith beneath the San Juans, active during late Oligocene time (28.2–26.5 Ma; see Table 1). One of these units, the Fish Canyon tuff, is one of the largest single ash-flows recognized in the world (~ 73,000 km^3 [8]).

2.2. Paleomagnetism

Several paleomagnetic studies have been conducted in the central San Juans, and the remanent magnetic directions reported [5,6]. The scatter that these authors observed in the data was attributed to uncertainties in tectonic corrections, to irregularities in the magnetic field at different sites including secular variation, and to field orientation problems. Paleopoles reported by these authors are $\phi = 298°E$, $\theta = 85.6°N$; $k = 24.1$ for 18 reliable sites [5] and $\phi = 114°E$, $\theta = 85°N$; $k = 11.32$ for 36 sites [6]. Using the results from the tuffs and other sites from western North America, Beck et al. [6] report a paleopole for the Oligocene of $\phi = 132.5°E$, $\theta = 81°N$.

A number of paleomagnetic analyses of other tuffs have been reported. For example, Grommé et al. [9] used the natural remanent magnetization (NRM) results as a correlation tool in several western North American tuffs, three of which are Oligocene in age. They found significant differences in NRM direction in successive cooling units that they attribute to uncertainties in making tectonic corrections. Reynolds [10] who examined some Yellowstone tuffs, also observed significant amounts of scatter in the remanent magnetic (RM) directions. He attributes this to a number of factors, including different cooling rates (varying blocking temperatures), variations in local magnetic anomalies, and a pronounced chemical overprint that appears to be greatest in the devitrified interiors of the tuffs. He suggests that magnetic anisotropy might be a potential source of dispersion, produced possibly during flow, but reports no AMS measurements.

Geissman [4] summarized much of the previous paleomagnetic work and the possible causes of RM dispersion, and demonstrates that small-scale flow and compaction-related movements of parts of a tuff, after a partial or complete RM acquisition, can produce significant errors in estimating a mean RM direction from a population of data points collected across the tuff. He demonstrates that reliable paleomagnetic directions can be obtained from the highly welded basal vitrophyre in tuffs.

Curie temperatures (T_c) were reported by Reynolds [10] and Geissman [4]. Both workers indicate that T_c in basal vitrophyres are consistent with those exhibited by relatively titanium-poor magnetities (540–580°C). It is interesting to note that the minimum estimated temperature necessary to produce a welded tuff is 600–625°C [11,12]. Some magnetic phases, in interior segments of the tuffs, exhibited $T_c > 580°C$ [4,10] and appear to be associated with chemical RM overprints possibly due to maghemitization.

2.3. Secondary mass flow (rheomorphism)

A number of authors have indicated that preferred orientations of fabric elements in tuffs may be related to secondary mass flow [7,11,13]. Wolff and Wright [7] use the term rheomorphism to refer to secondary mass flow that occurred shortly after emplacement while the magma was still a coherent viscous fluid. They examined peralkaline ignimbrites that exhibit high temperature secondary flow, and welded air-fall tuffs that exhibit secondary flow only after initial compaction, and suggest that rheomorphism is probably nearly ubiquitous in *peralkaline* welded tuffs deposited on slopes. Wolff and Wright [7] point out, however, that calc-alkaline rhyolite tuffs in the western

U.S. are generally higher in viscosity and may cool more rapidly during emplacement. Thus, rheomorphism may be inhibited in these tuffs by groundmass crystallization. An important point that Wolff and Wright [7] make is the presence of a linear fabric, which in the welded tuffs they examined, is interpreted to indicate that rheomorphism has occurred. Furthermore, the magnetic fabric in such tuffs is initially oblate and horizontal due to compaction. Rheomorphism produces a tendency towards the development of a prolate fabric.

2.4. Anisotropy of magnetic susceptibility (AMS)

Magnetic susceptibility (K or χ, volume or mass susceptibility, respectively) is a symmetrical second-rank tensor coefficient relating an induced magnetic moment to the inducing magnetic field [14]. It is commonly quoted in terms of susceptibility per unit volume or mass in SI units and therefore is a dimensionless Système International (SI) number (equivalent to MKSA Practical Rationalized units), or as gauss/oersted when using CGS Classical Unrationalized emu (see Hanna [15] and Payne [16] for a detailed discussion of MKS units used in magnetics, and Nye [14] for further elaboration in terms of magnetic susceptibility measurements).

AMS data for any rock specimen are conventionally expressed as triaxial ellipsoids whose major, intermediate, and minor axes represent the corresponding maximum (K_a), intermediate (K_b), and minimum (K_c) susceptibility magnitudes and azimuths (magnitude ellipsoid of Nye [14]). Bulk susceptibility can then be calculated from these magnitudes and is given as:

$$\bar{K} = (K_a + K_b + K_c)/3 \qquad (1)$$

For most rock specimens it has been shown that the shape of the total susceptibility ellipsoid expresses the preferred alignment of ferrimagnetic mineral grains [17,18] and therefore the AMS of such samples can be used as an indicator of petrofabrics in the rock [19].

It has been established that the AMS principal axes in rocks are *generally* controlled by the following [20,21]: (1) K_c (minimum) susceptibility azimuths are aligned normal to the plane of foliation in the rock (at the pole to foliation), (2) K_c axes align parallel with the axis of maximum compression (uni- or biaxial) in the rock, (3) K_a axes for turbulent grain suspensions generally align parallel with the direction of flow, (4) K_a (maximum) susceptibility azimuths parallel lineations in the rock and lie in the foliation plane, (5) K_a axes align normal to the direction of major compression, and (6) K_a axes are oriented parallel with the direction of maximum strain (extension). The above six effects result from such factors as flow alignment during and after emplacement, gravitational settling of particles, and post emplacement stresses such as compaction.

3. Methods

3.1. Field

Cores (106) were drilled with a gasoline-powered portable drill (MAC-35A) from 18 sites in roadcuts, and oriented using a Brunton compass (Fig. 1). Cores were separated laterally by an average distance of ~4 m. Four different tuffs were sampled. In stratigraphic order, from top to bottom, these are the Wason Park tuff (4 sites), the Carpenter Ridge tuff (3 sites), the Fish Canyon tuff (10 sites), and the Masonic Park tuff (4 sites). Sites 4–9 of the Fish Canyon Tuff were taken from a ~250 m vertical section of the tuff. Anomalous areas in the tuffs, such as sites within the source caldera, shear zones, highly weathered zones, vitric zones, etc., were not sampled. Specific sample locations are available from the author.

3.2. Laboratory

The cores of ~2.5 cm diameter, were sliced into lengths of ~2.2 cm to conform to an optimum length to diameter ratio range of 0.82–0.90 [22,23]. The AMS was measured for all samples from each site using an automated low-field torque magnetometer [24].

The NRM was measured for all samples using a Schonstedt SSM1A spinner magnetometer, coupled with a SWTP 6800 microprocessor. Pilot samples were partially alternating field (a.f.) demag-

netized, using a two-axis tumbler, at 10 mT steps to 90 mT and the RM measured after each run. Typical results were used in compiling the Zijderveld [25] vector plots in Fig. 2. Based on these results, the decision was made to perform a stepwise a.f. demagnetization at 10, 20, and 30 mT steps, and the RM data after 30 mT demagnetization were then used in compiling Table 1. Two samples from each tuff were given an isothermal RM in fields of 50, 100, 200, 300, and 600 mT and the magnetization measured. Thermomagnetic measurements were performed in vacuo with a digital thermomagnetic balance calibrated using synthetic magnetite, hematite, goethite, and maghemite powders provided by Toda Manufacturing Company in Japan.

3.3. Statistical

Since the AMS of the sample is a symmetric physical property, a principal AMS axis defines two directions on a unit sphere. A random scatter of the axial directions about some preferred azimuth defines a bimodal distribution of induced moment orientations. This is quite different from the unimodal scatter of RM vectors. Therefore, principal axis mean directions were calculated using the eigenvalue-eigenvector method (summarized, for example by Mark [26]) which is independent of any one direction assigned to a principal axis. The method entails calculating mean eigenvalues and eigenvectors using the summation matrix $\bar{A}$ for n observations, where:

$$\bar{A} = \sum_{i=1}^{n} \begin{vmatrix} x_i^2 & x_i y_i & x_i z_i \\ x_i y_i & y_i^2 & y_i z_i \\ x_i z_i & y_i z_i & z_i^2 \end{vmatrix} \quad (2)$$

and x_i, y_i, and z_i are the vector components for each individual sample principal AMS axis.

Within-site scatter about an AMS axial mean is represented by:

$$S_a \text{ or } S_c = (\lambda_1 + \lambda_2 + \lambda_3)/3 \quad (3)$$

and λ_1, λ_2, and λ_3 are the eigenvalues of the mean AMS azimuth. S_a or S_c can range from 1.0, representing a cluster of identical axial orientations and a unidirectional ellipsoid (highly prolate), to 0.333 for a random scatter of orientations whose representation quadric is a sphere. An example of how this procedure is employed is given by Ellwood and Abrams [21].

Standard Fisherian statistics [27] were used in calculating RM mean declinations and inclinations reported in Table 1, where:

$$k = (N-1)/(N-R) \quad (4)$$

is used as an estimate of precision.

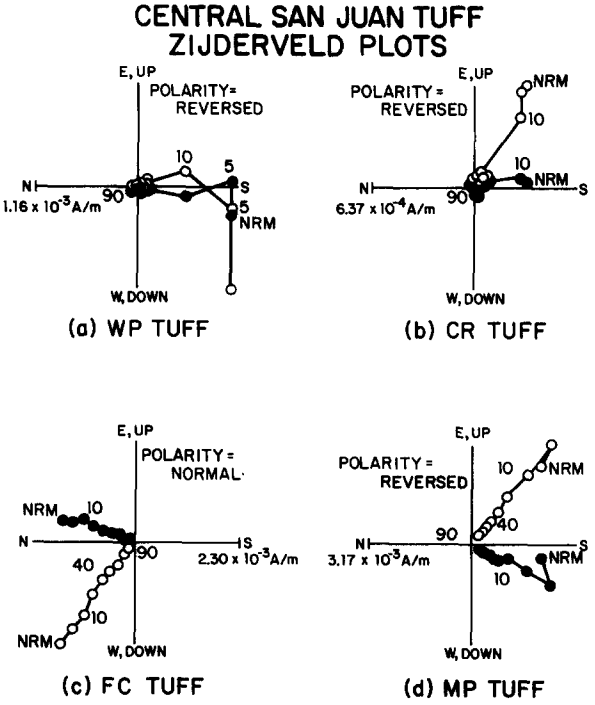


Fig. 2. Zijderveld [25] vector plot of RM directions from each tuff analyzed, after a.f. demagnetization to 90 mT. Filled circles represent projections on a horizontal plane. Open circles are plotted on a vertical plane. Intensity of magnetization in A/m.

4. Results

Mean RM directions for each site and for each tuff are reported in Table 1. Data from each site were evaluated for internal consistency, and two anomalous samples were eliminated from consideration in calculating site mean directions in Table 1. Two Fish Canyon tuff sites (FC-1 and FC-10, * in Table 1) included anomalous RM di-

TABLE 1

Remanent magnetic results

Site	N_1/N_2	Site means				Unit means
		D	I	$J\times10^{-3}$	k	
WP-1	6/6	165.0	−18.9	0.236	19.4	Wason Park tuff
WP-2	4/4	160.7	−22.7	0.359	405.2	$D=165.1$, $I=-22.1$; $k=270.1$
WP-3	6/5 **	165.1	−19.8	0.589	87.4	Age=~26.5 Ma
WP-4	6/5 **	169.7	−26.9	1.380	702.5	$\phi=284.1$, $\theta=-60.5$
CR-1	6/6	161.1	−60.1	0.915	305.9	Carpenter Ridge tuff
CR-2	6/6	156.7	−60.2	0.516	185.5	$D=149.0$; $I=-56.9$; $k=57.1$
CR-3	6/6	134.0	−48.4	21.667	118.5	Age=~27.5 Ma
						$\phi=352.3$, $\theta-65.6$
FC-1 *	6/6	24.1	6.3	0.274	13.9	
FC-2	6/6	359.6	53.9	0.151	413.2	
FC-4	6/6	12.1	54.5	0.435	141.6	
FC-5	3/3	24.0	46.6	0.264	962.8	
FC-6	3/3	35.4	52/3	0.344	2556.2	Fish Canyon tuff
FC-7	3/3	24.1	56.2	0.750	245.7	$D=16.2$; $I=53.7$; $k=117.8$
FC-8	3/3	13.3	54.0	1.022	107.2	Age=27.8 Ma
FC-9	3/3	5.4	55.9	0.844	492.1	$\phi=353.4$, $\theta=76.5$
FC-10 *	6/6	47.9	−1.7	0.405	115.1	
FC-11	6/6	13.4	52.0	0.945	799.4	
MP-1	6/6	202.4	−54.8	0.532	24.7	Masonic Park tuff
MP-2	6/6	195.2	−52.0	0.438	303.7	$D=180.6$; $I=-54.2$; $k=23.0$
MP-3	6/6	144.6	−44.9	0.013	41.4	Age=28.2 Ma
MP-4	6/6	185.5	−55.7	1.159	582.5	$\phi=172.4$, $\theta=-78.0$

Site location is from Fig. 1. N_1 = number of samples drilled at each site. N_2 = number of samples within each site used in calculating the site mean. D, I, and J = RM declination and inclination in degrees, and intensity of magnetization after a.f. demagnetization to 30 mT, in amperes per meter, respectively. k from equation (4) [27]. Age is from Lipman et al. [2]. ϕ = tuff paleopole longitude; θ = paleopole latitude. A site (FC-3) assumed when collected to be part of the Fish Canyon tuff, exhibited anomalous RM behavior (negative polarity) which is currently being investigated. Data for this site are not reported here.

* Sites rejected in calculating the tuff mean directions.

** Samples rejected in calculating the site mean directions.

rections and were not used in calculating the Fish Canyon mean direction.

Mean AMS data for each site are given in Table 2. Also reported are the precision indicators, S_a for K_a data and S_c for K_c data. The four typical AMS within-site data distributions present in these tuffs are presented in Fig. 3. Fig. 4 illustrates the excellent between-site and between-sample magnetic fabric consistency observed for the ~250 m thick Fun Valley stratigraphic section of the Fish Canyon tuff.

5. Discussion

5.1. Rock magnetism

Isothermal remanent magnetization (IRM) experiments for samples from each site were conducted to characterize the magnetite grain size distribution and contribution of hematite to the magnetization of these samples. Typical results are given in Fig. 5. Saturation or near saturation is reached in most samples at 0.2 T although a slight

TABLE 2

AMS site mean data

	N_1/N_2	K_a			K_c			%An	V	$\bar{K}\times10^{-2}$
		D	I	S_a	D	I	S_c			
WP-1	6/6	45.3	−7.0	0.958	15.5	80.6	0.995	1.3	61.9	0.38
WP-2	4/4	54.8	−24.0	0.977	7.5	58.7	0.943	1.5	49.9	0.47
WP-3	6/5	336.3	28.0	0.987	159.1	64.6	0.992	2.6	45.6	0.84
WP-4	6/5	78.9	6.8	0.870	246.0	86.3	0.994	1.2	64.2	1.88
CR-1	6/5	337.7	26.6	0.917	161.8	66.9	0.992	1.1	51.7	0.83
CR-2	6/4	347.6	35.5	0.984	190.8	52.5	0.974	3.5	46.0	0.14
CR-3	6/6	54.3	−12.2	0.911	338.9	55.6	0.960	2.0	66.5	0.22
FC-1	6/3	127.7	−30.5	0.993	54.7	26.5	0.989	1.4	61.2	1.77
FC-2	6/6	86.3	15.9	0.558	295.3	73.0	0.989	1.6	63.2	1.67
FC-4	6/5	119.9	−4.7	0.948	90.2	83.0	0.984	1.0	56.6	1.14
FC-5	3/3	104.6	−11.8	0.968	122.5	77.9	0.986	1.3	60.3	1.41
FC-6	3/3	108.4	3.6	0.887	233.0	88.1	0.960	1.5	51.5	2.05
FC-7	3/3	106.7	−12.6	0.921	119.1	77.6	0.996	1.2	65.5	2.41
FC-8	3/3	106.9	−5.0	0.975	119.8	84.5	0.991	1.2	71.1	2.24
FC-9	3/3	106.7	−9.2	0.954	135.6	79.1	0.995	1.3	67.6	2.50
FC-10	6/6	353.5	18.1	0.931	95.0	22.3	0.984	1.1	57.3	4.18
FC-11	6/6	59.7	18.3	0.973	233.6	69.9	0.994	1.4	65.3	2.94
MP-1	6/6	123.6	9.1	0.905	244.6	83.5	0.919	0.9	46.8	4.18
MP-2	6/6	316.4	12.9	0.896	168.3	75.8	0.992	0.7	60.3	2.65
MP-3	6/5	43.2	17.9	0.921	233.8	71.9	0.997	2.7	56.4	0.58
MP-4	6/4	45.0	−3.6	0.846	120.2	83.2	0.988	0.8	70.3	2.79

K_a = maximum; K_c = minimum AMS. $\bar{K}$ = bulk susceptibility. See text for an explanation of the precision of cluster parameter $S_{a,c}$ (1.0 = perfect cluster; 0.333 = random cluster). V is computed from the relationship $\sin^2 V = (K_b - K_c)/(K_a - K_c)$, and is an indicator of deformation [18]. Site, N, D, I, as in Table 1. $\%An = (K_a - K_c)/K_b \times 100$. Dashed lines delineate the Fun Valley stratigraphic section within the Fish Canyon tuff (see text).

increase is observed in two samples to 0.6 T. These data indicate that magnetite is the dominant magnetic carrier in all samples, although some slight contribution from another magnetic phase, probably the maghemite visible in polished section in some samples, does appear to be affecting the data. This is apparent from the thermomagnetic curves for the tuff samples (Fig. 6). Typical results are presented. An initial decrease upon heating, due to weight loss, is indicated for most samples and may be the loss of adsorbed water or water loss from clays produced during slight amounts of weathering. Curie hematite temperatures (T_c) range from ~525 to ~600°C, and generally exhibit a characteristic magnetite shape (Fig. 6). It is inferred that Curie temperatures greater than 580°C are the result of the partial maghematization of magnetite observed in those rocks [28]. Overall magnetite content, as indicated by variations in susceptibility, would appear to be variable (Table 2) since $\bar{K}$ ranges from 0.14 to 4.18×10^{-2}. IRM data show saturation over a wide range of intensities. The IRM saturation values and $\bar{K}$ for the same samples (Fig. 5) are directly proportional (linear regression yields a correlation coefficient (r) = 0.91, which is significant at >99.9% level) suggesting that both variables are mirroring the magnetite content, with the difference due, possibly, to variations in magnetite grain size.

RM vector plots, after demagnetization, exhibit two general trends. Intensity of magnetization in samples from the Wason Park (WP) and Carpenter

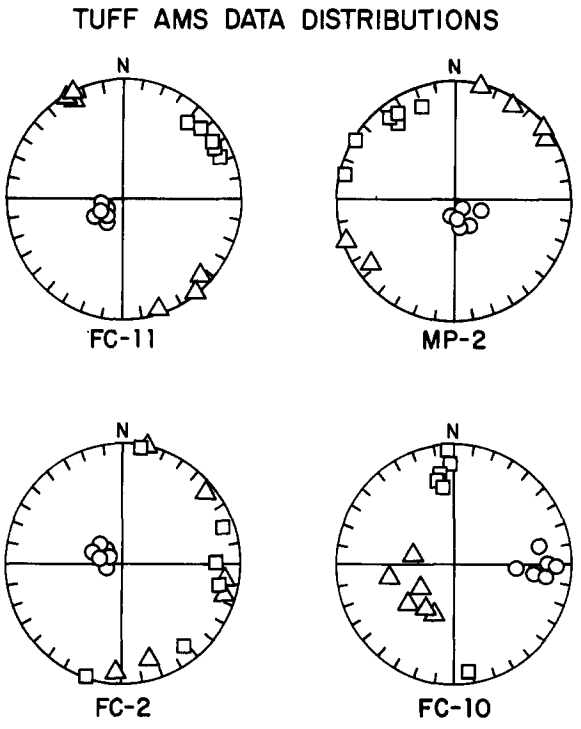


Fig. 3. Upper hemisphere, equal area plots of the four types of anisotropy of magnetic susceptibility (AMS) axial distributions observed for all sampled sites. □ = K_a (maximum); △ = K_b (intermediate); ○ = K_c (minimum).

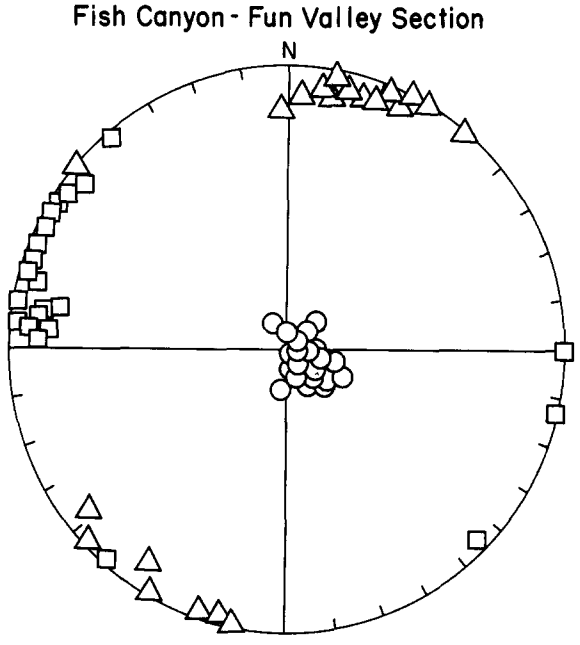


Fig. 4. Equal area, upper hemisphere plot of 21 samples from 6 sites, FC-4 to FC-9, taken from the 250 m thick Fun Valley section of the Fish Canyon tuff. Symbols as in Fig. 3.

Ridge (CR) tuffs (Fig. 5) decreases by an order of magnitude after only 20 mT a.f. demagnetization and exhibits the curving trends at low demagnetization fields characteristic of removal of a viscous RM (VRM) carried in relatively coarse-grained magnetite or low-coercivity maghemite. Samples change very little after 20 mT demagnetization, and trends are toward the origin. Samples from the Fish Canyon (FC) and Masonic Park (MP) tuffs (Fig. 5) generally show demagnetization behavior toward the origin in coherent steps. Some VRM behavior is observed, and NRM intensity is generally higher for these tuffs than for the Wason Park and Carpenter Ridge tuffs (Fig. 5), although intensity is generally similar in all tuffs after a.f. demagnetization to 30 mT (Table 1).

Interpretation of the vector plots, T_c measurements, $\bar{K}$, IRM data, and optical observations indicates that magnetite is present in all samples, as both coarse- and relatively fine-grained constituents. Some alteration to maghemite is apparent, reducing the effective grain size of the larger magnetite grains. The fine-grained magnetite is the stable RM carrier in these tuffs, with RM acquisition occurring during cooling between 525 and 580°C.

5.2. *Flow-direction estimates*

The magnetic lineation for each tuff site, determined from AMS measurements, was used to indicate the inferred axis of uniaxial flow (with respect to a known source; Fig. 1). The source can be located, if it is assumed that the axial directions from several sites within a single flow will intersect at the source of the ash-flow. The zones of these intersections are stippled in Fig. 1, and are located within or very near the source caldera for each tuff. One assumption in this method is that the linear fabric produced will be near horizontal (< 20°). Therefore source estimation for sites that exhibit anomalous, higher-angle linear fabric (e.g., WP-2, WP-3, CR-1, CR-2, FC-1, FC-2) is poor.

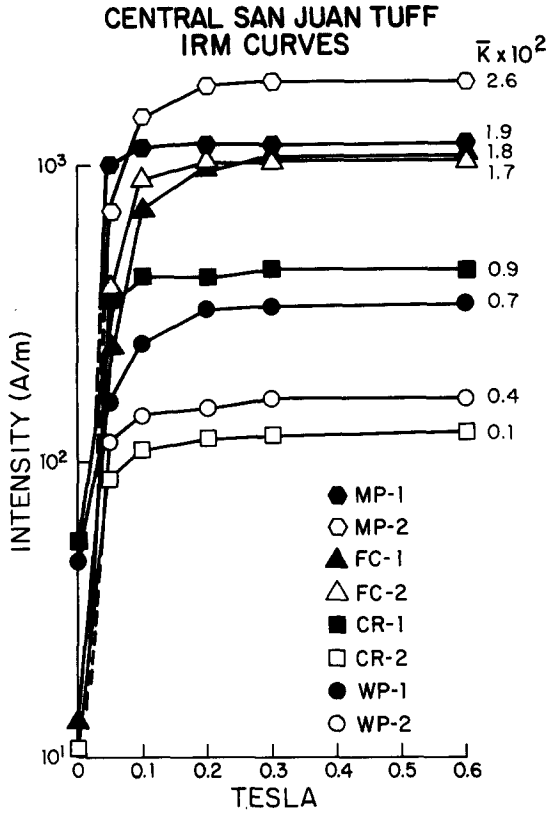


Fig. 5. Isothermal remanent magnetization (IRM) curves for two samples from each tuff analyzed.

The method involves measuring the AMS of individual samples from each site and determining the AMS for that site. Obviously anomalous samples were not included in determining the mean, and N_2 in Table 2 represents the number of samples from each site used for the final result. S_a (Table 2) is a measure of the precision of the K_a (maximum or long axis; in Figs. 3 and 4) cluster. S_a, therefore, is an indication of the precision about the mean direction ($K_a D$ and I, Table 2) of the linear fabric at that site in the tuff. A visual representation of how well clustered are the data for a given site is illustrated in Fig. 3. Site FC-11 (Fig. 3) has excellent precision, $S_a = 0.973$. Note that these data are not the best data (but are typical), as several sites in Table 2 yield higher S_a values. Precision for MP-2 (Fig. 3) is fair, $S_a = 0.896$, and a mean direction for this site is probably representative of the gross linear trend at the site. Also note that this site is typical of the worst data used in Fig. 1 as flow estimates. No distinct linear trend exists for site FC-2 (Fig. 3; dashed circle around the site location, Fig. 1). While lineation precision is excellent for samples from site FC-10 ($S_a = 0.931$), clearly the fabric at this site is anomalous (Fig. 3). Similar anomalous fabric is exhibited by site FC-1. These unusual sites are

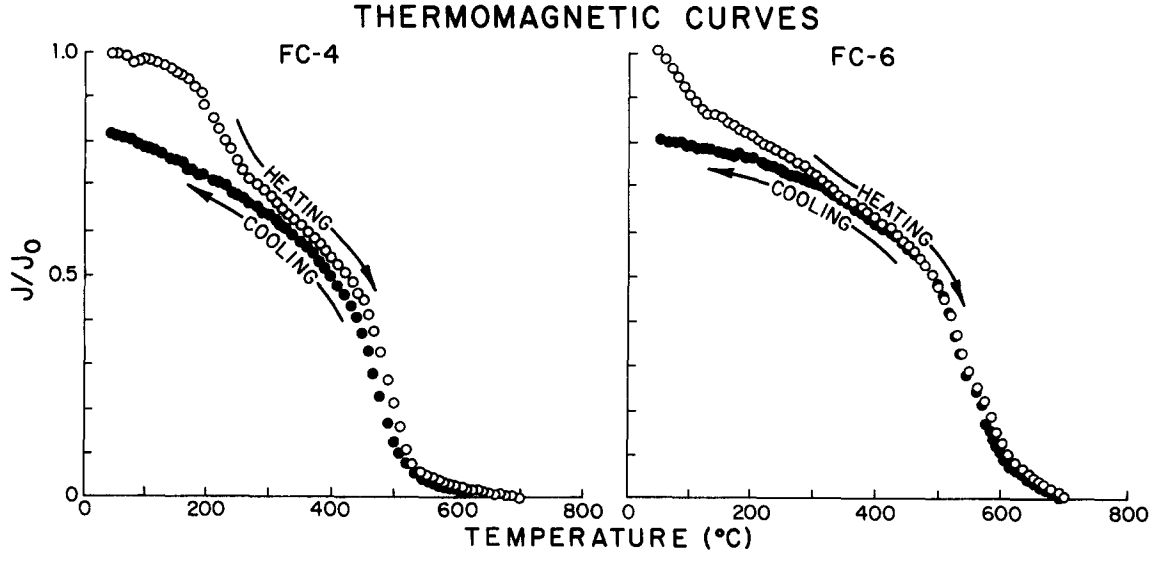


Fig. 6. Thermomagnetic curves for two samples from the Fish Canyon tuff. See text for discussion.

denoted in Fig. 1 by the dashed triangles around the site location. In Fig. 4 all the AMS data are presented from samples taken during a stratigraphic traverse of a ~250 m section of the Fish Canyon tuff. (Note that the one anomalous K_a direction (□ in Fig. 4) in the southwest quadrant of Fig. 4 is from site FC-4. This sample was not used in calculating the mean direction for the site). Fig. 4 illustrates the excellent intersite and intrasite consistency in the data, since flow margins (top and bottom) and flow interior are represented by discrete AMS axial clusters.

There is one point of magnetic fabric similarity for all tuff sites measured. The K_c (○ in Figs. 3 and 4) axes are tightly clustered within all but one site (MP-1) where the cluster is good (see values for S_c in Table 2).

5.3. Compaction

Because the K_c (minimum) principal axes develop parallel to the direction of maximum compressive stress (e.g. [18]), the data in Table 2 (K_c in Figs. 3 and 4) can give an estimate of compressive stress orientation during the period of magnetite grain mobility in the tuffs. Furthermore, it is expected that compaction of tuffs after emplacement should produce a near horizontal ($< 20°$ angle with a horizontal plane) magnetic foliation (defined by a great circle along which are distributed K_a and K_b principal axes (□ and △ in Figs. 3 and 4). Near-horizontal magnetic foliation is observed for most tuff sites. However, two sites exhibit high-angle ($> 50°$) foliation planes (FC-1 and FC-10) and are discussed below in the section on rheomorphism. Several sites (WP-2, WP-3, CR-1, CR-2, and CR-3) have foliation planes inclined at a moderate angle to the horizontal (20–40°). Exposures were not sufficient to establish if these sites had been tectonically tilted. However, the RM data for these anomalous sites (distributed over a large area) are very similar to those sites within the same tuff that exhibit a near-horizontal magnetic fabric (e.g., WP-1, and WP-3 in Table 1), suggesting that individual sites, if independently tilted after emplacement, have not been significantly rotated ($> 10°$). Furthermore, an RM tilt correction, using the AMS data and the assumption of an original horizontal magnetic foliation, did not improve the RM data consistency.

5.4. Rheomorphism

Two sites, well within the Fish Canyon tuff (FC-1 and FC-10), exhibit anomalous, high-angle magnetic foliation planes with K_c (minimum) clusters which are near horizontal. Tectonic displacement or tilting, which might be one explanation, must be discarded because site MP-3 in the Masonic Park tuff, which lies immediately below FC-10, has nearly horizontal magnetic foliation and is not tilted. Furthermore, the stratigraphic section containing these two sites exhibits no geological evidence in support of large-scale tilting. The preferred explanation for the formation of these high-angle magnetic foliation planes, is secondary, post-emplacement mass flow (rheomorphism). As Wolff and Wright [7] have indicated, calc-alkaline magmas have a tendency toward greater initial viscosities than exhibited by peralkaline magmas and this may be the reason that only two sites appear to have experienced rheomorphism. The vast volume and large thickness of the Fish Canyon tuff may have maintained high temperatures within the unit for long periods, maintaining lower viscosities in the interior of the ash-flow and thus allowing secondary flow. The magnetic fabric in sites FC-1 and FC-10 is unusual if it is argued that rheomorphism has produced a linear trend in the direction of secondary flow [7]. Rather, the fabric at these sites appears to be the result of simple shear [29], with the direction of strain oriented parallel with magnetic lineations. If this interpretation is correct, then a correspondence between the primary flow direction in this tuff and the orientation of linear fabric elements, which one might infer from site FC-1 in Fig. 1, is a geological accident. Therefore, flow direction estimates from such sites are suspect.

The question of whether or not rheomorphism has affected all sites, even though distinctly anomalous fabric does not exist at these sites, is difficult to answer. First, since reasonable RM precision exists between sites (Table 1), this suggests that rheomorphism is not totally disrupting the correspondence in RM directions, but rheo-

morphism may have a *slight* disrupting influence on the RM directions, producing some dispersion. Certainly the RM for sites FC-1 and FC-10, where rheomorphism has probably occurred, is anomalous (Table 1). Second, from an interpretation of the AMS data, slight magnitude rheomorphism may be indicated. It was pointed out by Wolff and Wright [7] that the petrofabric in tuffs is generally oblate in nature, and is modified by a tendency toward prolate shapes as a result of rheomorphism. Graham's [18] parameter, V, can be used as an indicator of oblate to prolate trends (Table 2). While the mean V for all sites indicates oblate shapes ($V > 45$), several sites fall very close to the prolate range ($V < 45$). Further, most of the sites fall significantly below the high V values (~ 70) expected for undeformed but compacted sediments [18]. The affect of slight magnitude rheomorphism may be to produce the observed scatter in K_a axial clusters (Table 2).

5.5. Remanent magnetism (RM)

Paleopoles have been calculated for the Carpenter Ridge, Fish Canyon, and Masonic Park tuffs and the results summarized by Beck et al. [6]. The results reported here (Table 1) are in general agreement with those reported by Beck et al. [6]. However, the number of sites reported here are insufficient to revise the paleopole reported by Beck et al. [6]. Between-site RM consistency is an important element of the data, since it indicates that the RM means for these central San Juan tuffs, are generally good indicators of the geomagnetic field direction at the time of, or shortly after (cooling through 525–580°C) emplacement of tuffs. (This assumes that a regional tectonic event has not tilted, rotated, or thrust the area en-bloc into its present position.) Therefore, the numerous possible elements which may tend to disrupt the data, such as tectonic tilting, temporal or spatial magnetic field irregularities, field orientation problems, chemical overprinting, deflection of RM vectors as a result of high percent anisotropy (observed here to be insignificant; low % An in Table 2), and rheomorphism, are either readily identifiable, and then eliminated from consideration (e.g., FC-1 and FC-10), or are random effects which cause slight increases in RM dispersion but do not significantly affect the mean RM direction in these tuffs.

The four tuffs, for which data are presented here, represent a continuous stratigraphic section that was formed during intermittent eruptions, 28.2–26.5 Ma. At least two polarity changes are inferred from the data. The first of these occurred between emplacement of the Masonic Park tuff and the Fish Canyon tuff. The second occurred between emplacement of the Fish Canyon tuff and the overlying Carpenter Ridge ruff.

6. Summary

Based on the remanent and induced magnetic properties determined for samples from four calc-alkaline ash-flow tuffs, located in the Central San Juan mountains of southwest Colorado, the following can be concluded:

(1) Flow direction estimates for individual sites using the anisotropy of magnetic susceptibility (AMS) method are reasonable. These are based on the assumption that linear magnetic elements in the flow, represented by the site mean long-axis (K_a) AMS data, parallel the flow direction.

(2) Eruptive source for individual tuffs can be identified as the point of intersection of several K_a AMS mean azimuths, projected from individual sites to source. While these intersections in this study occur in or near the known source calderas for these tuffs, it is apparent that at least three closely spaced intersections are necessary for reasonable estimates of an unknown source.

(3) Compaction and primary flow appear to be the dominant factors affecting the development of the magnetic fabric in these tuffs. Rheomorphism (secondary mass flow), however, dominates the fabric at two Fish Canyon tuff sites, producing anomalous, high-angle magnetic foliation planes, probably resulting from simple shear. The RM directions determined for these sites are also anomalous.

(4) A vertical section (6 sites) through ~ 250 m of the Fish Canyon tuff exhibits excellent between-site magnetic fabric and RM consistency.

(5) Paleomagnetic results for these tuffs [6] are

probably reliable indicators of the geomagnetic field direction in southwest Colorado during part of the Oligocene (28.2–26.5 Ma).

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